

Detection of Seizures Based on EEG Signals: A Machine Learning Approach

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Abstract—Epilepsy, a disorder that affects many people worldwide, is defined by recurring seizures, which can be monitored and detected with electroencephalography (EEG). Automating seizure detection is a challenging problem due to the fact that EEG signals are noisy and complex. In this paper, we aim to develop a machine learning framework for seizure detection using the CHB-MIT database that can distinguish between seizure and non-seizure states. After extracting time-domain, frequency-domain, and wavelet transform features to be inputs for the models, we found that SVM had an accuracy of 61% and a recall of 84%, while XGBoost performed better in terms of accuracy, 80% and recall, 85%.

Keywords—*electroencephalography, seizure, frequency bands analysis, classification, discrete wavelet transform, SVM, XGBoost*

I. INTRODUCTION

Epilepsy is a neurological disorder that is characterised by surges of electrical brain activity that result in recurring and unprovoked seizures. One of the most common neurological disorders in the world, epilepsy affects all different kinds of individuals, although it is more common in very young and very old people. The cause of the disorder cannot be confirmed for all epileptic patients, but some factors that could be connected with the condition are genetic influences, trauma to the head or brain, developmental conditions such as autism, and injuries before birth [1]. While epilepsy manifests differently from individual to individual, it is well-known as the “seizure disorder,” as those are its identifying symptoms.

Seizures are sudden interruptions of normal function in the brain that are caused by excessive and hypersynchronous discharge of neurons [2]. The discharge leads to a burst of electrical activity that can result in abnormal and involuntary movements, behaviors, sensations, and emotions [1]. There are two main types of seizures: focal and generalized. Focal seizures originate in only one area of the brain and are limited to one hemisphere. Generalized seizures, on the other hand, involve all areas of the brain and have a bilateral hemispheric onset [3].

Monitoring and detection of seizures can be achieved through the use of electroencephalography (EEG). EEG is a recording method that uses small electrodes placed on the scalp to measure electrical signals coming from the brain. Scalp-EEG is non-invasive, meaning that it does not require the electrodes to be inserted into the body [4]. This makes it a

low-cost and low-risk procedure, which is one of the reasons it is widely-used to examine various kinds of neural behavior.

There are a number of benefits of automated seizure detection using EEG signals. Most importantly, manual seizure recognition by a trained neurophysiologist requires a high amount of labor and time, which is not always possible in a clinical setting [5]. Having an automated framework through which models can distinguish between normal brain activity and epileptic seizure activity can optimize the efficiency of seizure detection by cutting down labor, time, and error costs.

Previous research conducted in this area has analyzed a wide range of various features and algorithms to help detect epileptic seizures using machine learning. Scalp-EEG recordings in particular contain a high amount of noise and are prone to artifacts caused by eye and muscle movements and power-line noise, and therefore need to undergo data preprocessing before they can be analyzed. Thus, principal component analysis (PCA) and independent component analysis (ICA) are popular methods used in EEG seizure detection algorithms during the preprocessing stage [6]. ICA, in particular, is used for artifact removal. This technique decomposes a signal into a linear combination of multiple statistically independent components representing different sources, which can then be analyzed to remove components that are likely to be EEG artifacts [17].

One of the common types of methods used in seizure prediction are time-domain. This involves the models differentiating between seizure and non-seizure time windows, as visually, the difference is often distinguishable. The types of features that characterize these methods come from EEG epochs, which are fixed-length segments of the brain signal. Analyzing epochs can give features such as amplitude difference and time separation, which can both be used to classify between seizure or non-seizure [6]. However, a drawback of using time-domain methods is that they are often patient-specific, because baseline brain activity tends to vary from patient to patient. Therefore, it is difficult to generalize time-domain methods to multiple individuals.

Frequency-domain methods often include analysis of the different frequency bands that brain waves consist of. The most common bands for EEG are alpha (α), beta (β), theta (θ), delta (δ), and gamma (γ). These rhythms are known to be associated with different states and brain regions. Alpha (8-12 Hz) rhythms typically increase in amplitude during a relaxed,

awake state, especially when the eyes are closed. Beta (12-30 Hz) rhythms are usually suppressed when voluntary movements occur, and are seen more in the frontal and central regions of the brain. Theta (4-8 Hz) rhythms are found during sleep and wakefulness, and might reflect abnormal activity during the latter. Delta (1-4 Hz) rhythms are known to be involved during deep sleep stages and are found more in the frontal regions of adult brains. Finally, gamma (30-50 Hz) rhythms have been associated with seizure onset phases as well as during visual stimulation [5]. Since the EEG recordings are typically in time-domain, fast Fourier transform methods are used to convert signals into the frequency domain for further analysis [7]. This method decomposes a signal into various frequency components, making it easier to examine a signal's different frequency bands.

Another set of features that can be extracted from EEG signals come from wavelet analysis. Discrete wave transform (DWT) is used to break down raw EEG data into mutually orthogonal frequency components, while also keeping time information [8]. In a 4-level DWT (Fig. 1), which is commonly used in EEG analysis, a signal is first decomposed into approximation and detail coefficients for two different frequency ranges. In the next level, the approximation coefficients are decomposed again into approximation and detail coefficients for smaller frequency ranges. This process occurs twice more, until there are 4 levels of detail coefficients (D1-D4) and the last level of approximation coefficients (A4) [9]. From these decomposed sub-bands, various features can be engineered.

Features from DWT analysis include some basic statistics, such as standard deviation, variance, and kurtosis. However, a more important feature that can be extracted from the DWT components is entropy. Previous research has shown that EEG signals are chaotic, and therefore, entropy measures can be used to represent and describe the data. Specifically for EEG, entropy measures the variability of the signal and its spread [10]. Therefore, since EEG signals during seizures are characterized by bursts of electrical activity, sub-band wavelet entropies can be used to discriminate between pre-ictal (before the seizure) and non-seizure states [6].

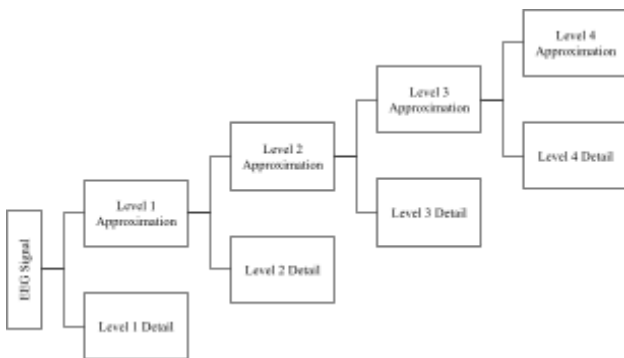


Fig. 1. Four-level DWT decomposition process. Adapted from [9].

In terms of binary classification algorithms used for seizure detection in EEG recordings, previous research identifies two main traditional machine learning ones: Random Forest (RF) and Support Vector Machine (SVM). RF

classifiers are ensemble methods that use a number of decision trees to classify each instance. The final decision is made by averaging the trees' probabilities of class assignments. SVMs use a hyperplane to create a separation between two classes, and try to maximize the margin between them [5].

SVMs utilize a kernel to transform the data inputted into a high-dimensional space, while then constructing an optimal hyperplane. A significant benefit of this approach is that it has good generalization, making it suitable for pattern recognition as is the case with seizure detection [18]. Prior research recognizes this method as one of the most commonly used for EEG-based seizure detection.

Another classifier that is used for seizure detection is the XGBoost classifier. XGBoost is an ensemble method that uses both decision trees and gradient boosting to classify data. A benefit of this algorithm is that it is computationally fast, which is important for detecting seizures with a large amount of EEG signals in a timely manner [15]. Implementations of feature importances and hyperparameter tuning also makes this algorithm favorable as EEG data typically result in a high amount of features.

There are few metrics with which seizure detection models can be evaluated with. The most simple is accuracy, which reveals how many epochs the model classified correctly as normal or preictal. Another significant metric is recall, or sensitivity, which measures how many preictal epochs the model actually classified as preictal. This is an important metric as it shows how good the model is at actually finding and detecting seizures. Finally, the F1 score balances recall with precision, which measures how many of the epochs that the model classified as preictal actually are. The F1 score takes both false positives and false negatives into account, making it a useful metric for examining the overall performance of a model [16].

In the present study, a seizure detection model using EEG signals has been developed. Data preprocessing and epoching around seizure events was performed first, followed by feature extraction to get model inputs. Based on prior research, two types of models were used to classify preictal (period before a seizure) versus normal epochs: SVM and XGBoost.

II. MATERIALS AND METHODS

A. Dataset Description

The data used in this project is the CHB-MIT Scalp EEG Database [11, 12, 13]. The EEG recordings in the dataset were collected from children with epilepsy at the Children's Hospital Boston. A total of 22 patients were recorded (5 males, 17 females), with ages ranging from 1.5 to 22 years. Subjects stopped anti-seizure medication a week before the data was collected.

The signals had a sampling frequency of 256 Hz (samples per second), and most recordings had a total of 23 signals, with a few having 24 or 26. The International 10-20 system (Fig. 2) determined the electrode positions and naming for this data, and a number of dummy signals were inserted in some files for easier reading. Finally, this database used bipolar

montages, meaning that each electrode was compared to its neighbor (i.e. Fp1-F7) to create a bipolar channel.

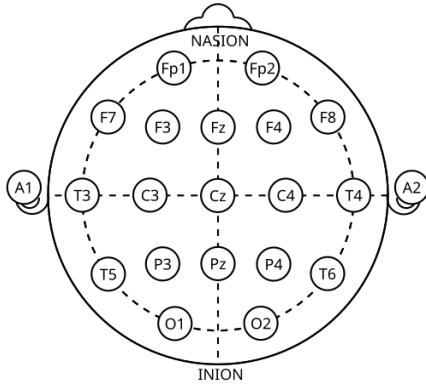


Fig. 2. The International 10-20 System for electrode positions [14].

B. Data Importing and Preprocessing

This database contains a large amount of EEG recordings data, with between 9 and 42 .edf (raw data) files per patient. Due to the extensive size of these files, only records with seizures were imported for examination. Between all of the subjects, this resulted in a total of 135 .edf files that were analyzed for this project.

Next, the seizure windows were extracted for all records by parsing summary text files [11], which identified the start and stop times for all seizures for a given case. These seizure windows were saved into a file for easy importing further on without requiring re-extraction.

The first step of data preprocessing was filtering. A notch filter was used to remove power-line noise at 60 Hz. In addition, a low-pass filter at 70 Hz was applied to reduce high-frequency noise and muscle artifacts. A high-pass filter at 0.1 Hz was also applied to reduce drifts in signals that might have been caused by perspiration and head movements.

Since the main focus of this project was detecting seizures, the next stage was epoching the raw data. Each file was segmented into 5 second chunks with no overlap. The pre-ictal state was defined as starting from 10 minutes before the seizure occurred, and the chunks within that time frame were labeled “preictal.” Then, data from another 10 minutes before that period was taken and labeled as “normal,” as long as it did not coincide with a seizure that had occurred previously.

Once all of the files were split into epochs around seizure events, the final preprocessing that was conducted was artifact removal. Autoreject and ICA were used for this process. The former is a Python library that is designed for automated artifact rejection, and was used to determine the threshold for noise and rejecting epochs that did not meet that threshold. Then, ICA was fit to the data and kurtosis was used to identify noisy components. Those components were then excluded from the data, resulting in epoched data with artifacts reduced and/or removed. This entire process was conducted for each of the 135 files in the dataset. Finally, all of the epoched data was saved as .fif files so they can be loaded in later without needing to rerun the whole preprocessing process.

C. Feature Extraction

First, before features could be extracted, EEG channels needed to be standardized across all of the files for all of the patients. Since there were inconsistencies with the number of EEG channels between subjects and sometimes between files, channels that were not present in all of the files were identified and subsequently removed in order to have a universal set of channels. This resulted in a total of 18 channels that were found in every file.

The first set of features extracted were in the time domain. For every file, three features were extracted: baseline activity (mean), signal variability (standard deviation), and asymmetry and sharp peaks (kurtosis). These features were calculated for each bipolar channel in each epoch.

Then, frequency-domain features were calculated. The power spectral densities using Welch’s method were calculated for each of the epochs, which show the strength of activity at each frequency in the data. Then, using the five frequency bands mentioned above (alpha, beta, theta, delta, and gamma), the power for each of the bands was calculated. A visual representation of this decomposition is shown in Fig. 3, which shows the average band power in the five frequency bands across the brain for normal epochs versus preictal epochs for one of the files in the dataset.

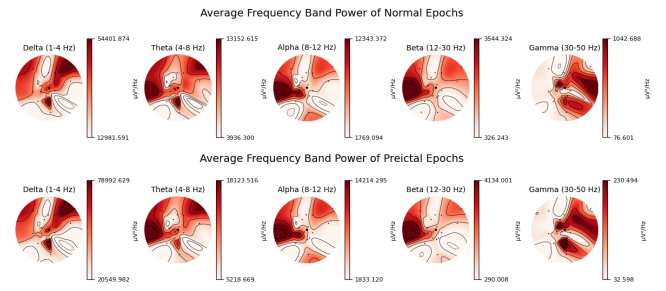


Fig. 3. Topographic maps comparing the average frequency band power for normal versus preictal epochs. Delta, theta, and alpha bands show an increase in power, while gamma power significantly decreased.

Finally, discrete wave transform features were the last set to be extracted from the data. First, the signal was decomposed into sub-bands using a four-level DWT and a Daubechies 4 wavelet (Fig. 4). For each of the sub-bands, standard deviation and kurtosis were extracted, as well as energy and relative energy in the band. Next, sub-band wavelet entropy was calculated, using the Shannon entropy equation. The last feature that was calculated was logarithmic band power, which has been commonly used in previous research in this area. Therefore, a total of six features were determined for each sub-band, and this process was repeated for every epoch.

After all of these features were calculated, the labels that identified whether the epoch was normal (0) or preictal (1) were extracted from the event IDs of the epoch. Then, all of the features were combined together into a single data frame and the labels were added as a column, resulting in a large dataset of all of the engineered features as well as a binary label column.

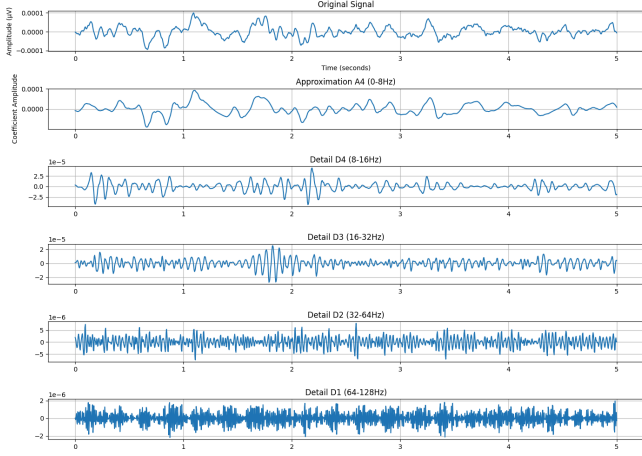


Fig. 4. Decomposition of EEG signal into sub-bands using DWT from lowest to highest frequency ranges.

D. Modeling

Before any models could be developed, a train-test split needed to be conducted. However, although we first tried splitting the data into a training patient set and a testing patient set, this ended up not the best choice since seizure activity did not appear to be generalizable between patients. Therefore, we split each patient’s individual data into training and testing sets that each model would be fit on and classify for.

The first model used was a Support Vector Machine. The kernel that was used for this model is the radial basis function (RBF), which is frequently used in more complex classification tasks since it can handle non-linear data and relationships, which is not possible with the linear kernel.

The second model that we implemented was the XGBoost Classifier. The initial model was simply the base model and did not include any changes to any hyperparameters of the classifier. This model was trained on each patient’s training set and tested on their testing set.

Next, hyperparameter tuning was implemented for the XGBoost model. Since we decided to build the model to be patient specific, we performed separate hyperparameter tuning for each patient.

Finally, we implemented two methods of feature selection: Variance Threshold and K-Best. Variance Threshold selects the minimum number of features needed to describe above a certain threshold of the variance in the data. K-Best selects the k most important features for capturing the dataset.

III. RESULTS

With the SVM algorithm, for the testing dataset, we saw an accuracy of 0.61, with a recall of 0.84. Precision was also slightly low, at 0.62, while the F1 score was 0.71. We found that the reason why there appears to be a large difference between the accuracy and the recall is that the SVM model is biased towards preictal classification, identifying a large proportion of epochs as preictal over normal.

For the base XGBoost model that did not have any hyperparameter tuning, accuracy did go up from SVM, at

0.78. All other metrics increased as well (Table 1), and there was less of a gap between accuracy and recall like seen in the SVM results. This reveals that the XGBoost model was better at classifying compared to SVM, and didn’t extremely favor either preictal or normal.

Incorporating the features selection done through Variance Threshold and K-Best, we tested the hyperparameter tuned model on the recommended subset of the data. For Variance Threshold features, that decreased the number of columns from 360 to 109 using a threshold of 0.005 – the given columns capture 99.5% of the variance. We had K-Best select the top 50 and top 100 features. All three versions performed worse than the original model, however, they were significantly faster. With Variance Threshold feature selection, the model ran 80% faster. With K-Best top 100 features, the model ran 81% faster. And, with K-Best top 50 features, the model ran 90% faster.

TABLE I. PERFORMANCE OF MODELS FOR SEIZURE DETECTION

Model	Metrics			
	Accuracy	Precision	Recall	F1 Score
SVM	0.61	0.62	0.84	0.71
XGBoost Initial	0.78	0.79	0.84	0.81
XGBoost Tuned	0.80	0.80	0.85	0.83
XGB and Variance Feature Selection	0.78	0.79	0.83	0.81
XGB and 50 KBest Feature Selection	0.70	0.72	0.77	0.75
XGB and 100 KBest Feature Selection	0.77	0.78	0.83	0.80

Once hyperparameter tuning was run for the XGBoost model, we saw an improvement when evaluating on the testing data sets. Most significantly, accuracy hit 0.80 with this model, and recall went up to 0.85. Precision and F1 score also increased, to 0.80 and 0.83, respectively. Of the three models that we tested for this data set, XGBoost with hyperparameter tuning had the best performance, and the lowest difference between accuracy and recall, meaning it wasn’t too biased towards the majority class of preictal.

IV. DISCUSSION

Our results demonstrate the capability of machine learning models to accurately predict the onset of seizures. Correctly identifying 85% of preictal windows is promising for the automation of seizure detection to ensure proper care can be given. However, there is certainly room for improvement and further experiments to see if decreasing the preictal window leads to higher detection accuracy.

In addition, feature extraction was a crucial stage in this project. Although many features were engineered, a number of them could be cut out with no consequences on performance, as shown during feature selection. Therefore, there is definitely room for more relevant features to be extracted from the EEG signals, and future work should explore more options in terms of new features that can improve detection.

Furthermore, one aspect of seizures that was not examined in this project was differentiating between focal and generalized seizures. Since these manifest differently in the brain, the features might not look the same for each, and distinguishing between them when detecting seizures could play a role in improving model performance.

When deciding between parameters for the XGBoost model, we chose to optimize recall without lowering the other metrics significantly. Recall is important to optimize because seizures are a medical emergency and a lack of warning could be extremely harmful. However, a high recall should not come at the expense of a low accuracy, in other words, the model predicting nearly every window to be preictal.

Latency, the time between first classification of the seizure by the model and the actual onset, is another important metric to take into consideration. While the tuned XGBoost model with all features had the highest accuracy, the time to train and test was significantly longer. XGBoost with Variance Threshold feature selection may be a better performing model due to its capability to quickly process and predict incoming data, since signals will be input in real-time. Further research into the average latency of the models in predicting would help differentiate and rank the effectiveness of each.

V. CONCLUSION

Seizures are often recognizable through fluctuations in brain signals long before the seizure occurs. Integrating machine learning algorithms can warn patients and doctors of the onset of a seizure long before visible symptoms occur. By developing an automated framework for detecting seizures, we can provide better treatment and avoid the labor intensive task of manual detection.

Support vector machines and XGBoost were both able to detect epileptic seizure activity in the brain from EEG signals, but the XGBoost classifier did outperform SVM. After feature selection and hyperparameter tuning, this performance increased even more. However, we believe that the incorporation and implementation of deep learning models could result in overall better seizure detection.

One limitation of this study was the size of the data. EEG recordings are large, and since recordings were of multiple hours, the data files were too big to work with on our personal computers. Therefore, segmentation was necessary during this project, but that meant the raw data could not be worked with as extensively. Another limitation was that this project focused on patient-specific seizure detection. Hence, it is difficult to generalize findings to all patients.

Using simple machine learning techniques in this project, we achieve a high accuracy, demonstrating the capabilities of seizure detection with technology. While a strong start, future work should focus on understanding the average latency of the model, additional feature engineering that may capture relevant changes in signals, and the implementation of deep learning algorithms for improved performance.

APPENDIX

This project was designed and written using Python in Jupyter Lab. The full code can be viewed on GitHub at the following link: [GitHub Repo](#)

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